

Inverse Kinematics of the Main Arm

We start with the diagrams of the arm, split into a top view and a side view

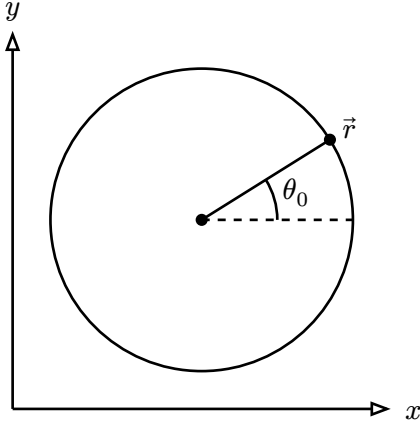


Figure 1: Top View

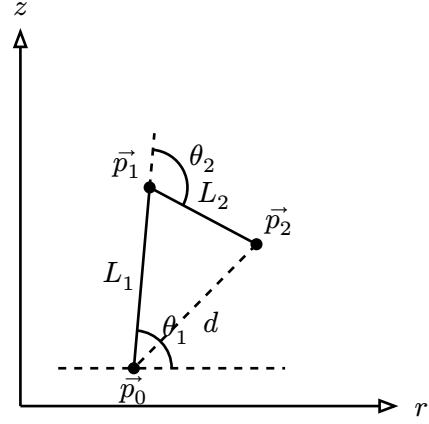


Figure 2: Side View

We already know L_1 and L_2 , as they are the lengths of the robot's arms which are given. $\vec{p}_0$ and $\vec{p}_2$ are also known as they are the position of the base which is static and the goal position of the end effector.

We are trying to solve for θ_0 , θ_1 and θ_2 , which are the joint angles the robot needs to reach the desired positions.

To solve this first notice that $\vec{p}_0$, $\vec{p}_1$, and $\vec{p}_2$ make a triangle. We also know all 3 side lengths so we can use the law of cosines to figure out each of the interior angles and then convert them into the angles we need with some more simple math.

Definition: The Law of Cosines - $c^2 = a^2 + b^2 - 2ab \cos(C)$ where C is the angle opposite side c

$$d = |\vec{p}_2 - \vec{p}_0| \quad (1)$$

To solve for θ_2

$$\begin{aligned} d^2 &= L_1^2 + L_2^2 - 2L_1L_2 \cos(\pi - \theta_2) \\ 2L_1L_2 \cos(\pi - \theta_2) &= L_1^2 + L_2^2 - d^2 \\ \cos(\pi - \theta_2) &= \frac{L_1^2 + L_2^2 - d^2}{2L_1L_2} \\ -\cos(\theta_2) &= \frac{L_1^2 + L_2^2 - d^2}{2L_1L_2} \\ \theta_2 &= \cos^{-1} \left(\frac{d^2 - L_1^2 - L_2^2}{2L_1L_2} \right) \end{aligned} \quad (2)$$

To solve for θ_1 we first solve for the interior angle of the triangle, opposite to L_2 , the same way as we did for θ_2 then to get the angle from the r -axis to $\vec{p}_2$, we can do $\text{atan2}(\vec{p}_2)$. Therefore

$$\theta_1 = \text{atan2}(\vec{p}_2) + \cos^{-1} \left(\frac{L_1^2 + d^2 - L_2^2}{2L_1d} \right) \quad (3)$$

And finally, to solve for θ_0 we can just do

$$\theta_0 = \text{atan2}(\vec{r}) \quad (4)$$

To put it all together, to find the joint angles needed to get the end effector to $\vec{p}_2$, we have the following equations

$$\begin{aligned} \theta_0 &= \text{atan2}(\vec{p}_{2_y}, \vec{p}_{2_x}) \\ \theta_1 &= \text{atan2}\left(\vec{p}_{2_z}, \sqrt{\vec{p}_{2_x}^2 + \vec{p}_{2_y}^2}\right) + \cos^{-1}\left(\frac{L_1^2 + |\vec{p}_2 - \vec{p}_0|^2 - L_2^2}{2L_1 |\vec{p}_2 - \vec{p}_0|}\right) \\ \theta_2 &= \cos^{-1}\left(\frac{|\vec{p}_2 - \vec{p}_0|^2 - L_1^2 - L_2^2}{2L_1 L_2}\right) \end{aligned} \quad (5)$$

Forward Kinematics of the Main Arm

forward kinematics of the arm can be calculated pretty easily, starting with 2D, we can just use basic trigonometry, making sure to convert the joint angles, into angles relative to the r -axis

$$\vec{p}_2 = L_1 \begin{bmatrix} \cos(\theta_1) \\ \sin(\theta_1) \end{bmatrix} + L_2 \begin{bmatrix} \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) \end{bmatrix} \quad (6)$$

To convert that from 2D to 3D we need to change the frame of reference from the rz plane to xyz space, and to do this we can just rotate r by θ_0 to go from the r -axis to the xy plane, resulting in this as the final equations for forward kinematics

$$\vec{p}_2 = \begin{bmatrix} [L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)] \cos(\theta_0) \\ [L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)] \sin(\theta_0) \\ L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \end{bmatrix} \quad (7)$$

Forward Kinematics of the Stylus

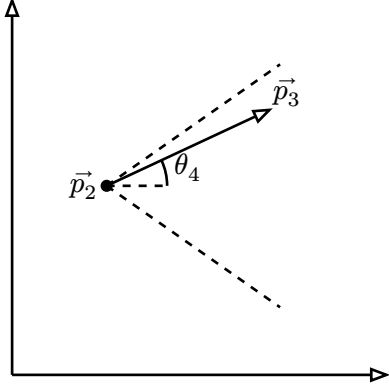


Figure 3: Side View of the Stylus

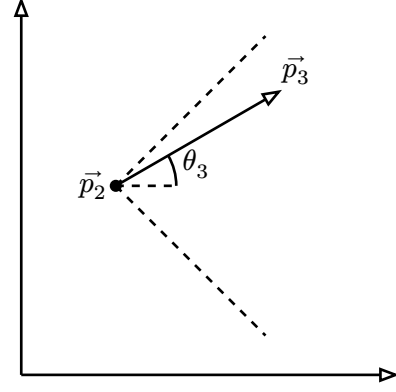


Figure 4: Top View of the Stylus

The kinematics of the stylus were given in the documentation, so copying it here:

$$\varphi = \theta_1 + \theta_2 \quad (8)$$

$$\mathbf{R}_{\text{head}} = \begin{bmatrix} \cos(\varphi) \cos(\theta_0) & \cos(\varphi) \sin(\theta_0) & \sin(\varphi) \\ -\sin(\theta_0) & \cos(\theta_0) & 0 \\ -\sin(\varphi) \sin(\theta_0) & -\sin(\varphi) \cos(\theta_0) & \cos(\varphi) \end{bmatrix} \quad (9)$$

$$\vec{u}_{\text{stylus}} = \begin{bmatrix} \cos(\theta_4) \cos(\theta_3) \\ \cos(\theta_4) \sin(\theta_3) \\ \sin(\theta_4) \end{bmatrix} \quad (10)$$

$$\vec{u}_{\text{world}} = \mathbf{R}_{\text{head}} \vec{u}_{\text{stylus}} \quad (11)$$

$$\vec{p}_3 = \vec{p}_2 + r \vec{u}_{\text{world}} \quad (12)$$

where r is the distance in the direction of the stylus from the head

Inverse Kinematics of the Stylus

To invert the forward kinematics all we have to do is solve for $\vec{u}_{\text{stylus}}$ given $\theta_0, \theta_1, \theta_2$ and $\vec{p}_3$

Starting from equation Equation 12 we notice that $\vec{u}_{\text{world}}$ is normal so to get rid of r we can just normalize

$$\vec{u}_{\text{world}} = \frac{\vec{p}_3 - \vec{p}_2}{\|\vec{p}_3 - \vec{p}_2\|} \quad (13)$$

And then using equation Equation 11 we can invert $\mathbf{R}_{\text{head}}$ to get

$$\vec{u}_{\text{stylus}} = \mathbf{R}_{\text{head}}^{-1} \vec{u}_{\text{world}} \quad (14)$$

And then converting $\vec{u}_{\text{stylus}}$ is just a matter of converting from rectangular coordinates to spherical

$$\theta_3 = \arctan\left(\frac{\vec{u}_{\text{stylus}} \cdot \mathbf{y}}{\vec{u}_{\text{stylus}} \cdot \mathbf{x}}\right) \quad (15)$$

$$\theta_4 = \arcsin(z)$$

Now the only difficult part is solving for $\mathbf{R}_{\text{head}}^{-1}$, however we don't need to do this as it is more efficient to solve the linear equation directly rather than use the inverse and matrix multiply.

Especially since after entering it into a symbolic solver, the mathematical inverse is extremely long and unwieldy.